

Nickel-Catalyzed Reductive Cross-Couplings: New Opportunities for Carbon–Carbon Bond Formations through Photochemistry and Electrochemistry

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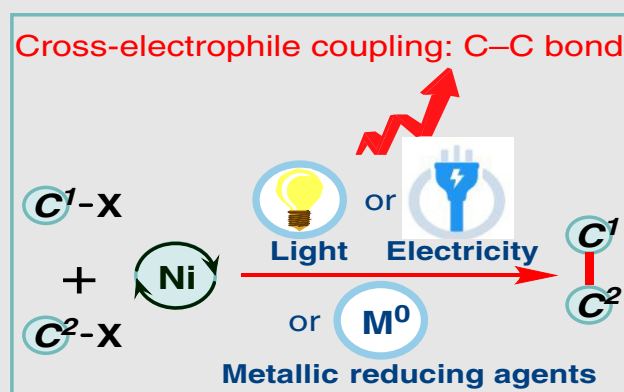
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Metal-catalyzed cross-electrophile couplings have become a valuable tool for carbon–carbon bond formation. This minireview provides a comprehensive overview of the recent developments in the topical field of cross-electrophile couplings, provides explanations of the current state-of-the-art, and highlights new opportunities arising in the emerging fields of photoredox catalysis and electrochemistry.



Reductive nickel catalysis displays unique features compared with the traditional methods and indicates a high potential for cross-electrophile couplings. These reactions provide an efficient and environmentally friendly way for the construction of C–C bonds.

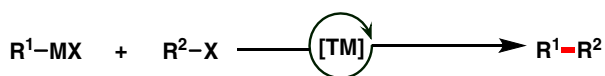
Keywords: reductive cross-couplings, photochemistry, electrochemistry, nickel catalysis, carbon–carbon formations

Introduction

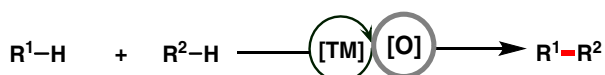
Carbon-carbon bond formations have always been one of the most useful reactions in both industry and academia and have gained considerable attention from many synthetic chemists who developed novel strategies to achieve improved and sustainable transformations. Transition metal catalysis has continually provided new activation modes for C-C bond formations¹⁻⁵ and fascinated synthetic chemists for a long time. Many named reactions have been associated with transition metal catalysis and become a powerful method for the cross-couplings of electrophiles with organometallic nucleophiles (Scheme 1a). Despite this progress, the use of organometallic reagents can cause undesired side reactions and chemical wastes. Alternatively, cross-nucleophile coupling has been developed as an efficient method for the synthesis of synthetically and biologically important compounds (Scheme 1b).⁶⁻⁸ However, the lower availability of carbon nucleophiles represents a limitation. Recently, transition metal-catalyzed cross-coupling reactions between two bench stable electrophiles under reductive conditions have emerged as a powerful tool for the construction of C-C bonds. In particular, nickel (Ni) catalysts, characterized by low reduction potential and electro-negativity, can undergo rapid oxidative addition.⁹ As such, it is not surprising that nickel-catalyzed reductive cross-coupling represents a flourishing area in organic chemistry with characteristic advantages over classical synthesis, such as widely available carbon electrophiles and avoiding unstable organometallic reagents and time-consuming and costly prefunctional processes.

Thus, there has been significant progress in the development of new cross-electrophile couplings for constructing C-C bonds. The first example of cross-electrophile coupling was published about 100 years ago by Wurtz¹⁰ and Tollens and Fittig¹¹ using sodium metal as the reductant and mediator for the cross-coupling of aryl halides with alkyl halides. Stoichiometric sodium metal and high temperatures are needed. Therefore, the functional group tolerance and application of this method are limited. Another classical cross-electrophile coupling strategy is electrosynthesis. Early explorations of electroreductive nickel-catalyzed cross-coupling include the cross/homo-coupling of organic halides, acyl, and carboxylation reductive cross-couplings.¹² However, electroreductive cross-couplings can be difficult to achieve and specialized laboratory equipment is required. These limitations have restricted the further development of this method for the formation of C-C bonds for several years. However, electrosynthesis has recently seen a renaissance and emerged as a useful tool for reductive cross-couplings. The most popular strategy of reductive cross-coupling reactions is the combination of nickel catalysis and metallic reducing agents and number of nickel-catalyzed cross-electrophile couplings have been developed using Mn or Zn as reductants.¹³⁻²⁰ Despite its success, in addition to scalability and efficiency problems, the utility of metal powders inevitably produces excess waste. Thus, photochemical alternatives for reductive cross-coupling reactions have been developed. Recently, impressive achievements have been made by merging nickel catalysis with photo- and electro-chemistry to create new activation modes and avoid the use of metal powders. Considering the advantages of nickel-catalyzed cross-electrophile coupling for C-C bond construction, it is important to provide a conceptual understanding of this emerging area (Scheme 1c). Against this background, we attempt to give an overview of the current state-of-the-art of cross-electrophile couplings and highlight the new developments of reductive catalysis pathways.

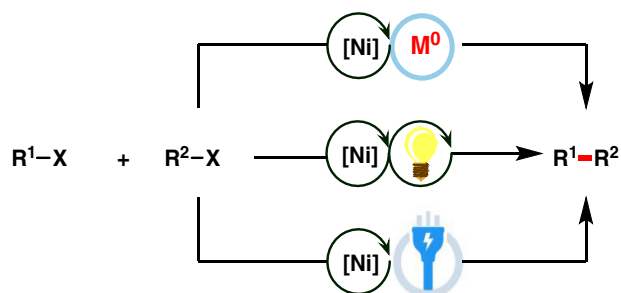
(a) Traditional cross-coupling



(b) Oxidative cross-coupling



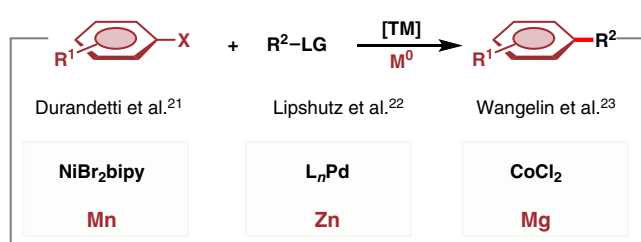
(c) Reductive cross-coupling



Scheme 1 | (a-c) Transition metal-catalyzed cross-coupling.

Alkyl-Aryl Reductive Cross-Coupling Nickel/metallic reducing agent system

The viability of alkyl-aryl cross-coupling via the combination of transition metal catalysis and metallic reducing agents was initially demonstrated by the research groups of Durandetti,²¹ Lipshutz,²² and Wangelin²³ (Scheme 2). Specifically, Durandetti and co-workers²¹ described an efficient nickel-catalyzed cross-coupling of aryl halides and α -chloroesters, as well as the Reformatsky reaction in the presence of manganese as the reducing metal. Lipshutz and co-workers²² investigated the participation of zinc in palladium-catalyzed cross-coupling of alkyl halide and aryl bromide, and Wangelin and co-workers²³ reported cobalt-catalyzed cross-coupling of



Scheme 2 | Early examples reporting the use of metallic reducing agents in transition metal catalysis.

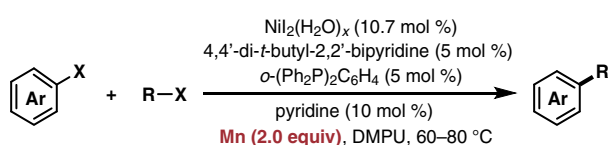
aryl and alkyl halides. These early examples demonstrated the potential of the combined transition metal/reducing metal systems as a useful tool to construct C-C bonds under milder conditions.

More recently, reductive cross-electrophile coupling was more widely recognized as a general concept and emerged as an actively researched and exciting area of transition metal catalysis. In 2010, Weix and co-workers²⁴ developed a Ni/Mn system for the selective cross-coupling of equimolar quantities of alkyl and aryl halide. High cross-selectivities were achieved using both a bipyridyl and a phosphine ligand (Scheme 3a).

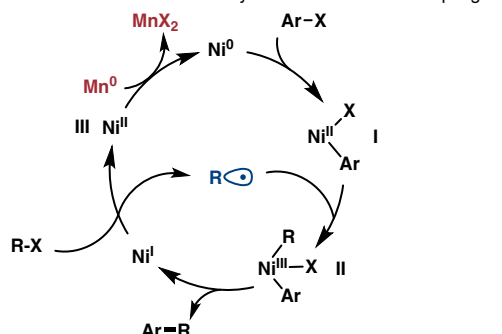
In the developed protocol, stoichiometric organometallic reagents were not required, and a broad range

of alkyl and aryl halides were tolerated. As a drawback, the use of secondary alkyl bromides resulted in mixed isomer products. Nevertheless, the direct reductive cross-coupling of alkyl halides with aryl halides without the formation of intermediate organomanganese species demonstrated the utility of this synthetic protocol. Regarding the mechanism (Scheme 3b), it is postulated that in the first key step the selective oxidative addition of aryl halides to low valent Ni(0) generates the Ar-Ni(II) intermediate **I**. Subsequent radical addition affords the Ar-Ni(III)-R species **II**. Finally, the reductive elimination of **II** affords the desired product and generates a Ni(I) species **III**, which could produce the alkyl radical from the alkyl halides via single-electron transfer (SET) or halogen-atom abstraction. Reduction of the Ni(I) species **III** by Mn finishes the catalytic cycle. Concurrently, similar results using cobalt/phosphine catalytic systems were disclosed by Amatore and Gosmini²⁵ with electron-deficient aryl bromides. After these studies, great efforts from many research groups have focused on the different alkyl electrophiles for nickel-catalyzed reductive cross-couplings (Scheme 3c).^{26–38} Notably, Molander and co-workers^{39,40} successfully expanded this strategy for the installation of alkyl fragments onto pharmaceutically relevant heterocyclic motifs. A variety of secondary aliphatic halides or tosylates underwent the reductive cross-coupling reaction in moderate to good yields, furnishing substituted

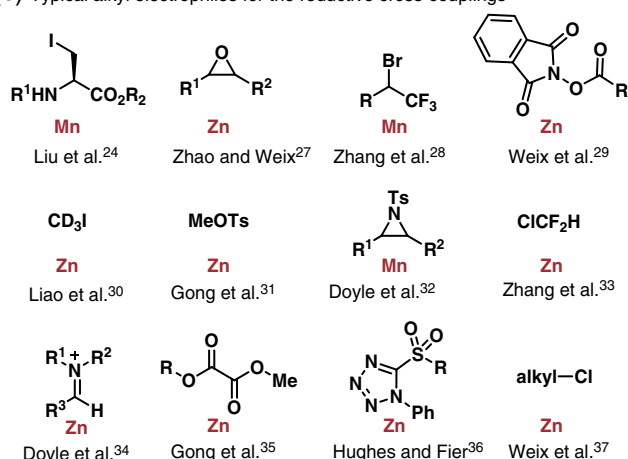
(a) Reductive cross-coupling of aryl halides with alkyl halides by Weix et al.²⁴



(b) General mechanism for the nickel-catalyzed reductive cross-couplings



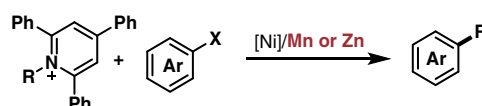
(c) Typical alkyl electrophiles for the reductive cross-couplings



Molander et al.^{39,40}



Rueping et al.⁴², Watson et al.⁴³, Martin et al.⁴⁴, and Han et al.⁴⁵



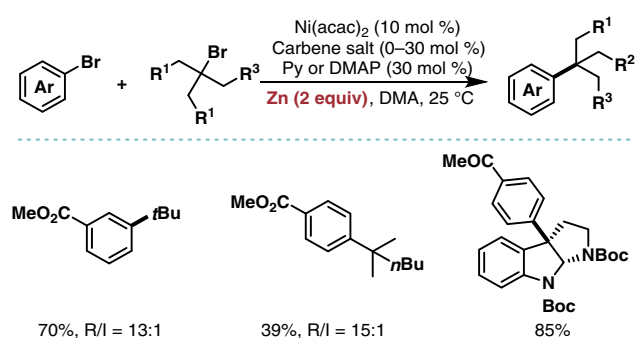
Scheme 3 | (a–c) Overview of nickel-catalyzed alkyl-aryl reductive cross-couplings.

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Scheme 4 | Reductive cross-coupling of aryl bromides with tertiary alkyl halides.

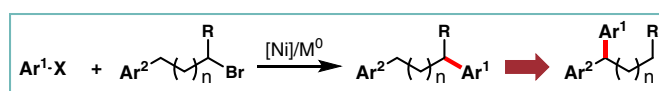
heteroaromatic compounds. Despite these achievements, the use of alkylamines, which are abundant natural feedstocks, had not been realized until more recently. In 2017, Watson and co-workers⁴¹ first reported the Suzuki-Miyaura cross-coupling of aryl boronic acids, employing Katritzky salts as C-centered-radical precursors. Very recently, the groups of Rueping,⁴² Watson,⁴³ Martin,⁴⁴ and Han⁴⁵ independently applied Katritzky salts in nickel-catalyzed reductive cross-couplings of aryl halides. In these cases, Mn and Zn were employed as the optimal stoichiometric reductants and elevated temperatures were usually required. Notably, Han's Ni/Zn system enabled wider substrate scope including bromoalkynes and alkyl bromides.

Although the nickel-catalyzed reductive cross-couplings of aryl halides with primary or secondary electrophiles are well developed, the use of tertiary alkyl electrophiles in nickel catalysis is not easy due to the dominant β -hydride elimination side reaction. In 2015, Gong and co-workers⁴⁶ resolved the issue by using pyridine (Py) or 4-(*N,N*-dimethylamino)pyridine (DMAP) and a carbene salt as the additives to suppress β -hydride elimination and enhance coupling efficiency (Scheme 4). The reaction tolerated various substituted aryl bromides and better results were obtained with electron-withdrawing substituents.

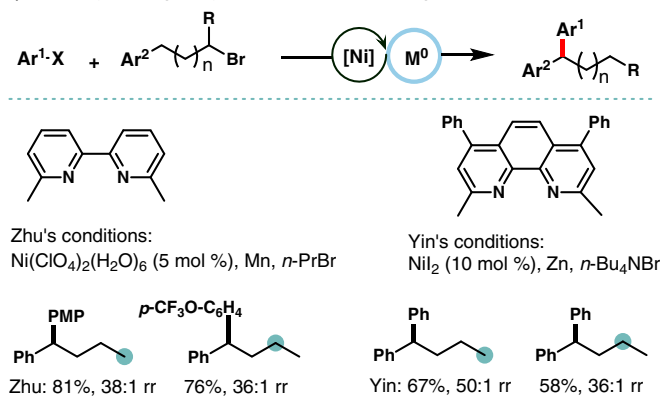
Until now, the cases described form C-C bonds at the ipso-carbon of alkyl halides, where the regioselectivity of these reductive cross-couplings has been less explored (Scheme 5a). An early example of migratory reductive cross-coupling was reported by Zhu and co-workers⁴⁷ in 2017 (Scheme 5b). The reaction proceeded smoothly using $\text{Ni}(\text{ClO}_4)_2(\text{H}_2\text{O})_6/6,6'$ -dimethyl-2,2'-bipyridyl as the catalyst with different nonactivated alkyl electrophiles and aryl bromides affording 1,1-diaryllkane derivatives, which are widespread in natural products and biologically active molecules, in good to excellent yields and regioselectivity.

A proposed mechanism for this transformation is provided in Scheme 5c. Initially, the oxidative addition of $\text{Ni}(\text{O})$ with the inactivated alkyl bromide leads to $\text{Ni}(\text{II})$ complex I. Reduction of $\text{Ni}(\text{II})$ by Mn and the following β -hydride elimination and migratory insertion steps deliver the thermodynamically stable benzylic- $\text{Ni}(\text{I})$ species III.

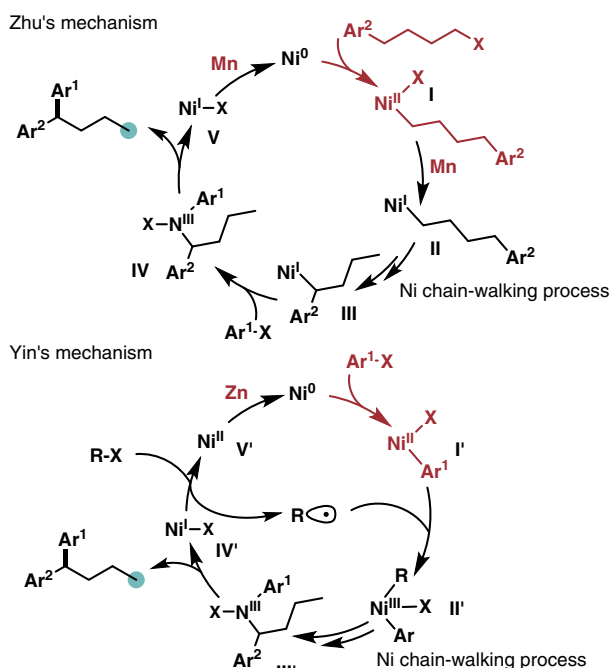
(a) Regioselectivity of the reductive cross-couplings



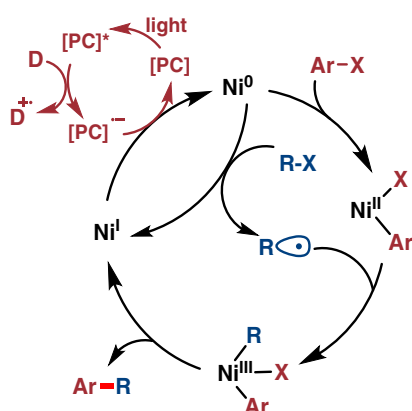
(b) Ni-catalyzed migratory reductive cross-couplings



(c) Proposed mechanism



Scheme 5 | (a-c) Nickel-catalyzed migratory reductive cross-couplings.



Scheme 6 | A general representation of reductive cross-coupling pathway via nickel/photoredox dual catalysis.

Then, the oxidative addition of the aryl bromides with Ni(I) generates the Ni(III) complex **IV**. Finally, the reductive elimination of Ni(III) complex provides the desired product and Ni(I)-X species **V**. The Ni(I)-X species is then reduced by Mn powder to close the catalytic cycle.

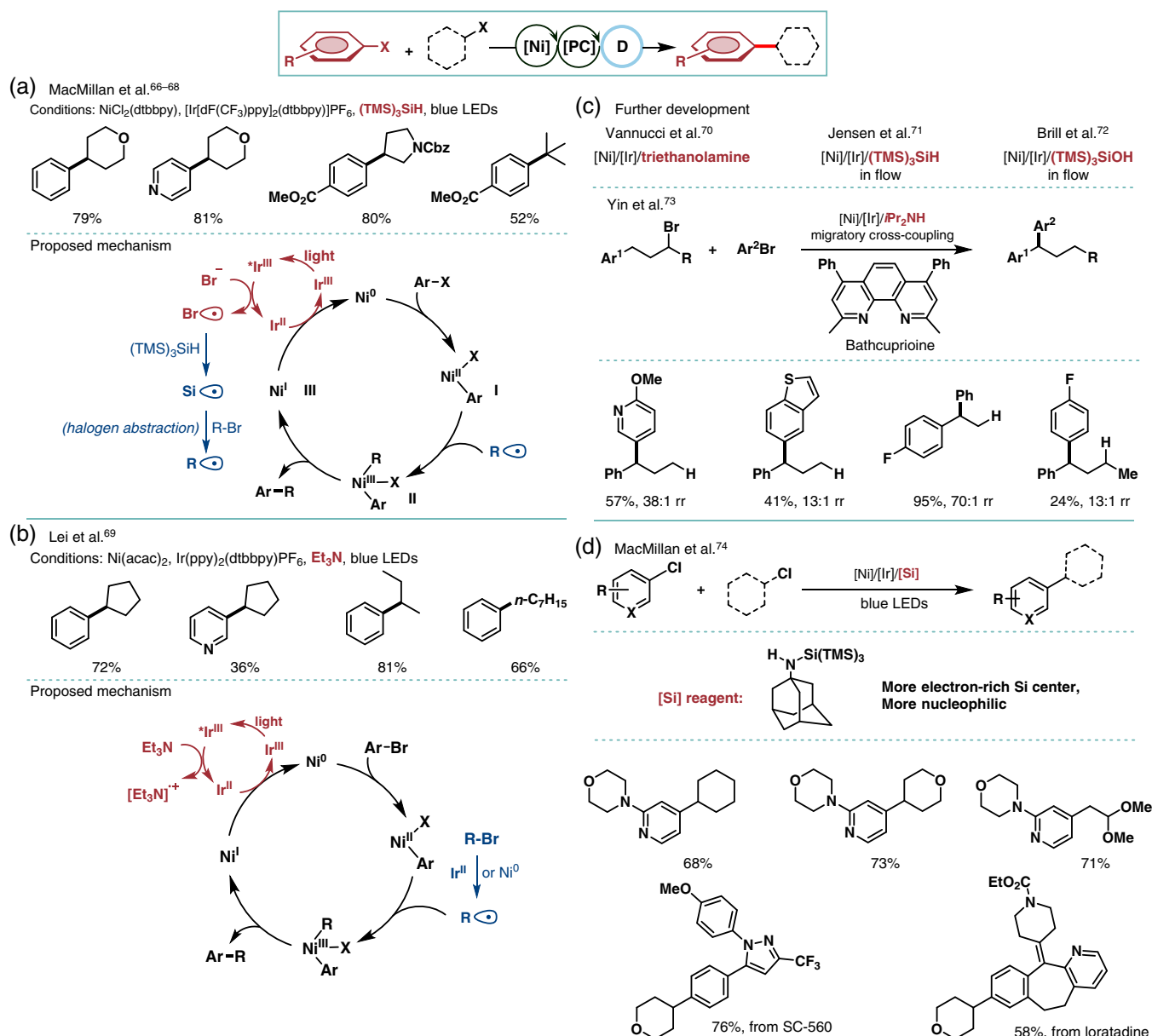
This class of transformation was also explored by Yin and co-workers,⁴⁸ who demonstrated the utility of NiI₂/bathocuproine as the cross-coupling catalyst and Zn as the reductant. Interestingly, an opposite mechanism is proposed. In the first key step, selective oxidative addition of aryl halides to Ni(0) rather than inactivated alkyl halides affords the Ar-Ni(II) complex **I'**. A radical chain process is proposed for the formation of the Ni(III) species **II'** through the addition of alkyl radical generated from inactivated alkyl halides via a SET of Ni(I)-X (**IV'**). Several control experiments and radical trapping experiments were carried out to support their mechanism. Although significant progress has been made in nickel-catalyzed reductive cross-coupling of aryl halides with alkyl electrophiles, the use of metal powders inevitably produces excess stoichiometric metal waste. Thus, the development of new types of reductive cross-couplings, which enable new activation modes, is still highly desirable. During the last few years, metal/photoredox dual catalysis has witnessed remarkable achievements and offered a broad range of unconventional transformations.^{49–65} To date, this field is strongly dominated by a redox neutral pathway, wherein the photoredox catalyst generates a radical species from the nucleophile partner and changes the oxidation state of the nickel catalyst for further transformations. A recently developed reductive nickel/photoredox system offers a valuable alternative for the construction of C-C bonds by using two electrophiles in the absence of a stoichiometric metal reductant (Scheme 6).

Nickel-photoredox dual catalysis

In 2016, MacMillan and co-workers^{66–68} developed the first nickel/photoredox dual catalyzed reductive cross-coupling of aryl bromides and alkyl bromides in the absence of metal powder (Scheme 7a). This reaction proceeded smoothly with a broad scope of alkyl bromides and aryl bromides through Ni/photoredox dual catalysis. Regarding the mechanism, it is postulated that in the first step the oxidative addition of aryl bromide to Ni(0) results in the formation of Ni(II) complex **I**. Concomitantly, a hydrogen-atom abstraction of tris(trimethylsilyl)silane (TTMSS) via bromine radicals forms a stabilized silyl radical intermediate. The following halogen-atom abstraction mediated by silyl radical generates the alkyl radical, which binds to the Ni(II) complex **I**, leading to the formation of Ni(III) intermediate **II**. Reductive elimination from **II** affords the desired product and Ni(I) species **III**, which can be reduced by the reduced photo complex Ir(II). In this case, the photoexcited catalyst is reduced by bromide to generate a bromine radical.

Lei and co-workers,⁶⁹ independently from the studies of MacMillan, reported an alternative strategy employing Et₃N as the terminal reductant. A proposed mechanism for this transformation is provided in Scheme 7b.

Initially, the oxidative addition of Ni(0) with aryl bromides leads to the Ni(II) complex. At the same time, the alkyl radical is generated via a SET of photoexcited catalyst or a low-valent nickel intermediate. The resulting alkyl radical is then intercepted by Ni(II) to form the Ni(III) species. Finally, the reductive elimination of Ni(III) species provides the desired product and Ni(I)-X species, which is reduced by the Ir(II) to close both catalytic cycles. This strategy was also used by Vannucci and co-workers,⁷⁰ who disclosed triethanolamine as the terminal reductant. Based on these previous developments, Jensen and co-workers⁷¹ developed a continuously stirred-tank reactor platform for the nickel/photoredox dual catalyzed cross-electrophile couplings in flow. Notably, gram-scale synthesis can be achieved after using their system for 13 h, which opened up potential applications of this system. A related approach was developed by Brill and co-workers⁷² for the assembly of complex drug-like compounds via cross-coupling of benzylic chlorides and (hetero)aryl bromides by using continuous flow and highlighting the industrial applicability. Furthermore, Yin and co-workers⁷³ successfully expanded this strategy to reductive migratory cross-coupling of alkyl bromides and aryl bromides by using bathocuproine as a nickel ligand (Scheme 7c). Compared with alkyl bromides and iodides, simple alkyl chlorides are abundant, inexpensive, and readily available through simple synthetic methods. As such, they are widely used as electrophilic partners in nucleophilic and aromatic substitutions. The concept of nickel/photoredox dual activation modes was also successfully expanded to the coupling of inactivated alkyl chlorides. The reaction involving the



Scheme 7 | (a–d) Overview of nickel/photoredox dual catalyzed reductive cross-couplings of aryl halides.

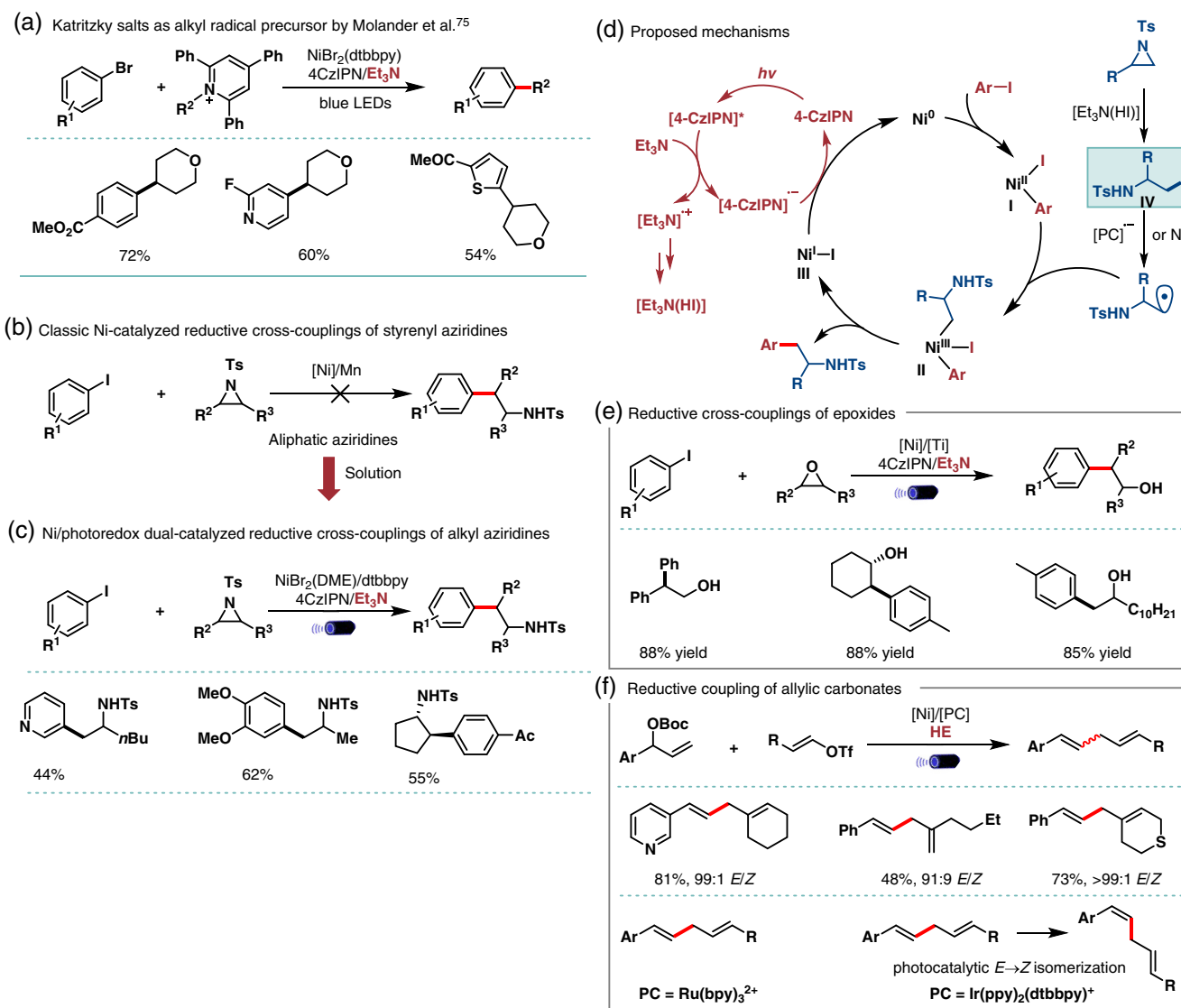
inactivated alkyl chlorides and aryl chlorides as the substrates, aminosilane as the reductant, catalyzed by $\text{NiCl}_2(\text{bim})$ and Ir-based photocatalyst, afforded the $\text{C}(\text{sp}^2)\text{--}\text{C}(\text{sp}^3)$ coupled products in generally good yields (Scheme 7d).⁷⁴ In the context of the development of alkyl electrophiles, a series of new nickel/photoredox dual catalyzed reductive cross-coupling reactions were developed using Katritzky salts, aziridines, and epoxides. In recent research by Molander and co-workers,⁷⁵ Katritzky salts have been identified as $\text{C}(\text{sp}^3)$ electrophiles for the construction of $\text{C}(\text{sp}^2)\text{--}\text{C}(\text{sp}^3)$ bonds. This reaction proceeded with good yields in the presence of 4CzIPN as the photocatalyst and $\text{NiBr}_2(\text{DME})/4,4'$ -di-*tert*-butylbipyridine (dtbbpy) as the cross-coupling catalyst, with a broad scope of differently substituted substrates (Scheme 8a).

In 2017, Doyle and co-workers³² developed a nickel/Mn-catalyzed reductive cross-coupling of styrenyl aziridines and aryl iodides. A drawback in the substrate scope was that an aliphatic aziridine did not work under Mn conditions (Scheme 8b). A recent study by Doyle and co-workers⁷⁶ constituted further progress in this field and resulted in a reductive nickel/photoredox strategy. Their study provided a new way for the reductive cross-coupling of alkyl aziridines and (hetero)aryl iodides. The newly developed system showed a broad substrate scope. A variety of substituted aryl iodides and alkyl aziridines underwent reductive cross-coupling to provide the desired products in moderate to good yields by using 4CzIPN as the photocatalyst and $\text{NiBr}_2(\text{DME})/\text{dtbbpy}$ as the cross-coupling catalyst. Notably, they were

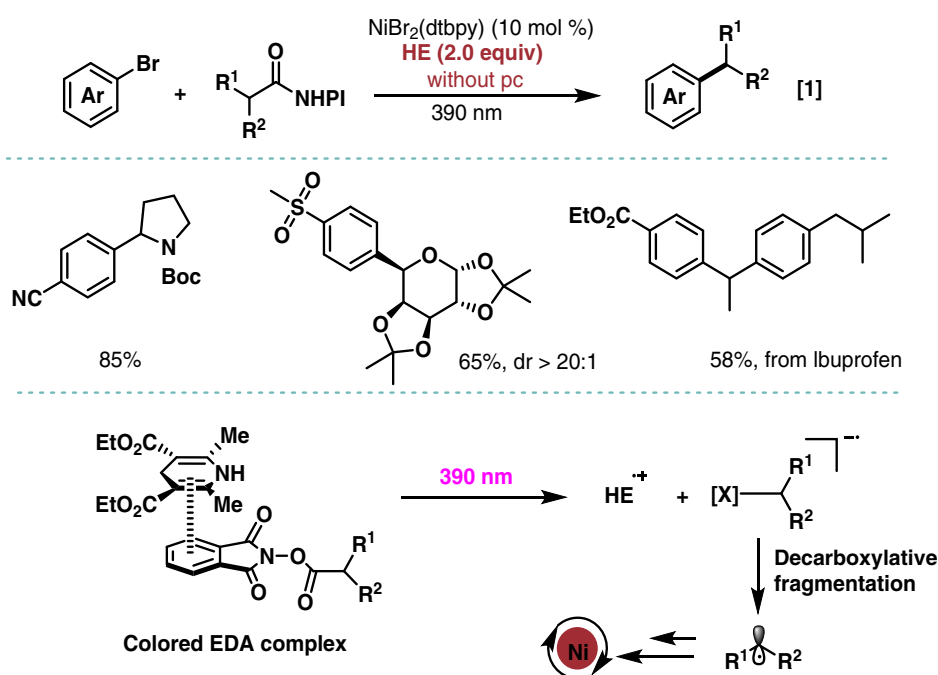
able to use cyclic aziridines that could not be employed in classic methods, which opened up potential applications of this strategy (Scheme 8c).

A proposed mechanism for this transformation is provided in Scheme 8d. Initially, the oxidative addition of Ni(0) with aryl iodide leads to Ni(II) complex **I**. The key β -iodoamine **IV** is formed via iodide ring-opening of aziridine. Subsequently, the alkyl radical is generated from β -iodoamine via a SET of photoexcited catalyst 4CzIPN^{•+} or a low-valent nickel intermediate or halogen-atom abstraction from Ni(I)–I (**III**). The resulting alkyl radical is then intercepted by **I** to form Ni(III) species **II**. Finally, the reductive elimination of **II** provides the desired product and Ni(I)–I species **III**. Then Ni(I)–I species **III** is reduced by the reduced state of [4-CzIPN]^{•–} to close both catalytic cycles. Notably, the classic nickel/Mn system gave the MnI₂ salt instead of β -iodoamine; thus, no desired products were obtained. Continued efforts by Doyle and

co-workers⁷⁷ expanded this strategy to cross-coupling of epoxides and (hetero)aryl iodides (Scheme 8e). Compared with the classic methods, the newly developed Ni/Ti/photoredox system has a wider substrate scope. Various styrene oxides, cyclic epoxides, and terminal aliphatic epoxides all reacted well to provide the desired products in moderate to good yields and high regioselectivities. Allylic carbonates have also been proven to be suitable electrophiles for reductive cross-couplings.^{78–80} Another nice example of cross-electrophile coupling via nickel/photoredox dual catalysis was described by Chu and co-workers⁸¹ by employing allylic carbonates and vinyl triflates as the substrates (Scheme 8f). In this case, both *E*- and *Z*-configured 1,4-dienes could be achieved by the choice of photocatalysts. When Ir(ppy)₂(dtbbpy)⁺ was used as the photocatalyst, a photoinduced contra-thermodynamic *E*→*Z* isomerization process would occur to give the (*Z*)-1,4-diene product.⁸¹



Scheme 8 | (a–f) The development of alkyl electrophiles via nickel/photoredox dual catalysis.



Despite the utility of these strategies, all of them rely on the redox potentials of the photocatalysts to furnish the cycle. In addition, many of these reactions become less appealing when considering the potential toxicity and high cost of the photocatalysts. The photoactive electron-donor-acceptor (EDA) complex allows for the generation of alkyl radicals under mild conditions without using metal based-photocatalysts and organic dyes. In the context of the development of milder nickel/photoredox strategy, Molander and co-workers⁸² reported the application of the EDA concept in nickel/photoredox dual catalyzed reductive cross-coupling of aryl bromides and *N*-hydroxyphthalimide (NHPI) esters. The reaction proceeds via the formation of an EDA complex between NHPI

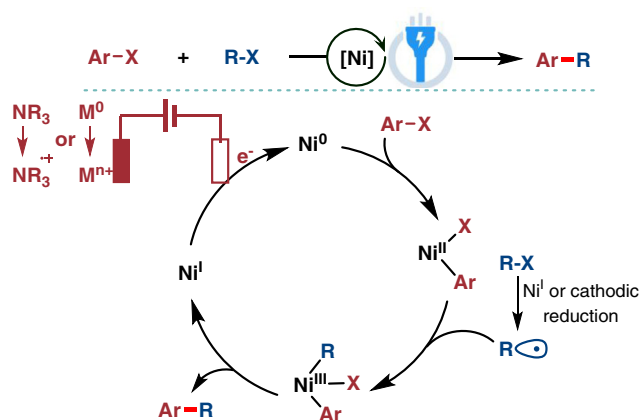
ester and Hantzsch ester (HE), which upon radiation-induced SET forms the alkyl radical (eq 1).⁸²

Electrochemical nickel catalysis

Electrochemistry may offer an economical and efficient alternative to classical transformations. Recent studies have demonstrated the ability of electrochemistry for various synthetically important bond-forming reactions. Within this research area, electrochemically induced metal catalysis has emerged as a powerful tool by integrating electrosynthesis and metal catalysis (Scheme 9). The seminal work of nickel-catalyzed reductive cross-coupling reaction between chloroesters and aryl halides was developed by an electrochemical strategy.¹² A breakthrough for nickel-catalyzed electroreductive cross-coupling as a general concept had not been achieved until very recently. In 2017, Hansen and co-workers⁸³ reported the nickel-catalyzed electroreductive cross-coupling of alkyl bromides with aryl bromides (Scheme 10a).

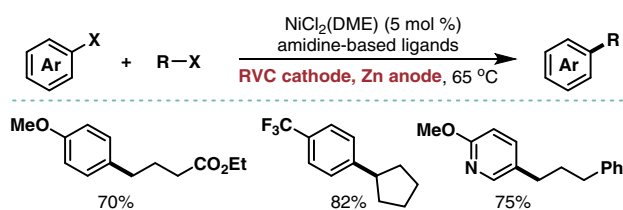
In this case, the sacrificial zinc anode was employed as the reductant and tuning of the current was found to be crucial to the reaction. The cooperative combination of nickel catalysis and electrochemistry circumvented the need of stoichiometric metal powders and high temperatures. Thus, this method exhibited good functional group tolerance and substrate generality.

Further efforts by Bio and co-workers⁸⁴ resulted in a more economical strategy. Compared with Hansen's method, the newly developed strategy used NHPI esters as the radical source and tertiary amine as the reductant under mild conditions in a divided electrochemical cell (Scheme 10b). Later, a one-pot electroreductive

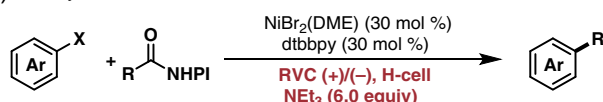


Scheme 9 | A general representation of electroreductive cross-coupling.

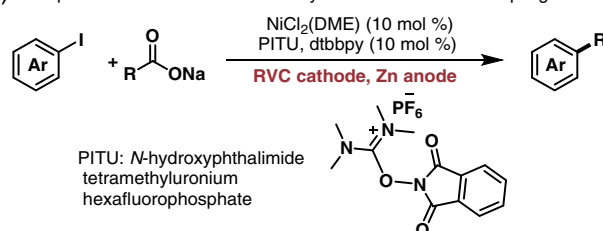
(a) Electrochemical nickel-catalyzed reductive cross-couplings



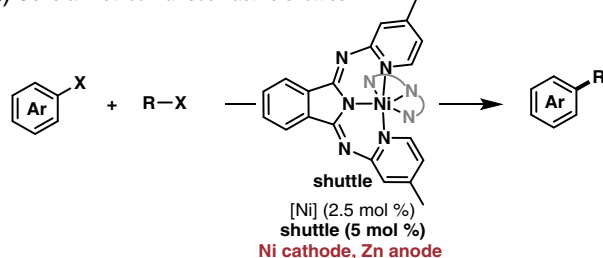
(b) Tertiary amine as the reductant



(c) One-pot electrochemical nickel-catalyzed reductive cross-couplings



(d) General method via redox-active shuttles

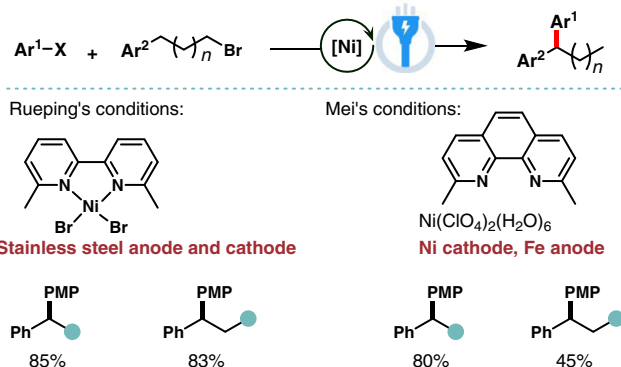


Scheme 10 | (a–d) Nickel-catalyzed electroreductive aryl-alkyl cross-couplings.

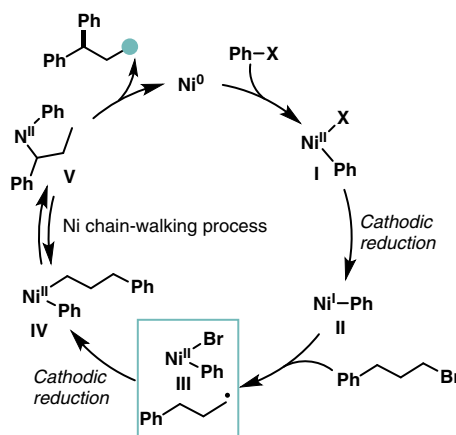
C(sp²)-C(sp³) cross-coupling reaction was developed by Loren and co-workers⁸⁵ (Scheme 10c), wherein the redox-active esters were generated in situ from alkyl carboxylates and NHPI tetramethyluronium hexafluorophosphate. Sevov and co-workers⁸⁶ developed a more general strategy and found that the redox-active shuttles can protect the nickel catalyst from over reduction, thus improving the yields and suppressing the side-product pathway (Scheme 10d). The reactions gave good yields across a wide range of substrates, including aryl, heteroaryl, or vinyl bromides with primary or secondary alkyl bromides.

Given the importance of 1,1-diaryllalkane derivatives, independently and concurrently, the research groups of Rueping⁸⁷ and Mei⁸⁸ reported the migratory electroreductive cross-coupling of alkyl halides and aryl halides (Scheme 11a). Both methods showed good functional group tolerance and wide substrate scope, giving rise to the corresponding 1,1-diaryllalkane derivatives in moderate to good yields. Based on the experimental results and density functional theory (DFT) studies, Rueping and

(a) Nickel-catalyzed migratory electroreductive cross-couplings



(b) Proposed mechanism



Scheme 11 | (a and b) Nickel-catalyzed migratory electroreductive cross-couplings.

co-workers⁸⁷ proposed a plausible mechanism for this transformation (Scheme 11b).

In the first step, the Ni(0) species is generated at the cathode surface. Subsequent oxidative addition of Ni(0) to aryl bromide gives Ni(II) complex I, which is reduced at the cathode giving rise to the Ar-Ni(I) intermediate II. A SET then occurs to generate alkyl radical and Ar-Ni(II)-Br species III. The following single-electron cathodic reduction and alkyl radical addition will give the Ni(II) intermediate IV. The β-hydride elimination and migratory insertion steps deliver the thermodynamically stable benzylic-Ni(II) species V. Finally, the reductive elimination of benzylic-Ni(II) species would release the desired product and regenerate the Ni(0).

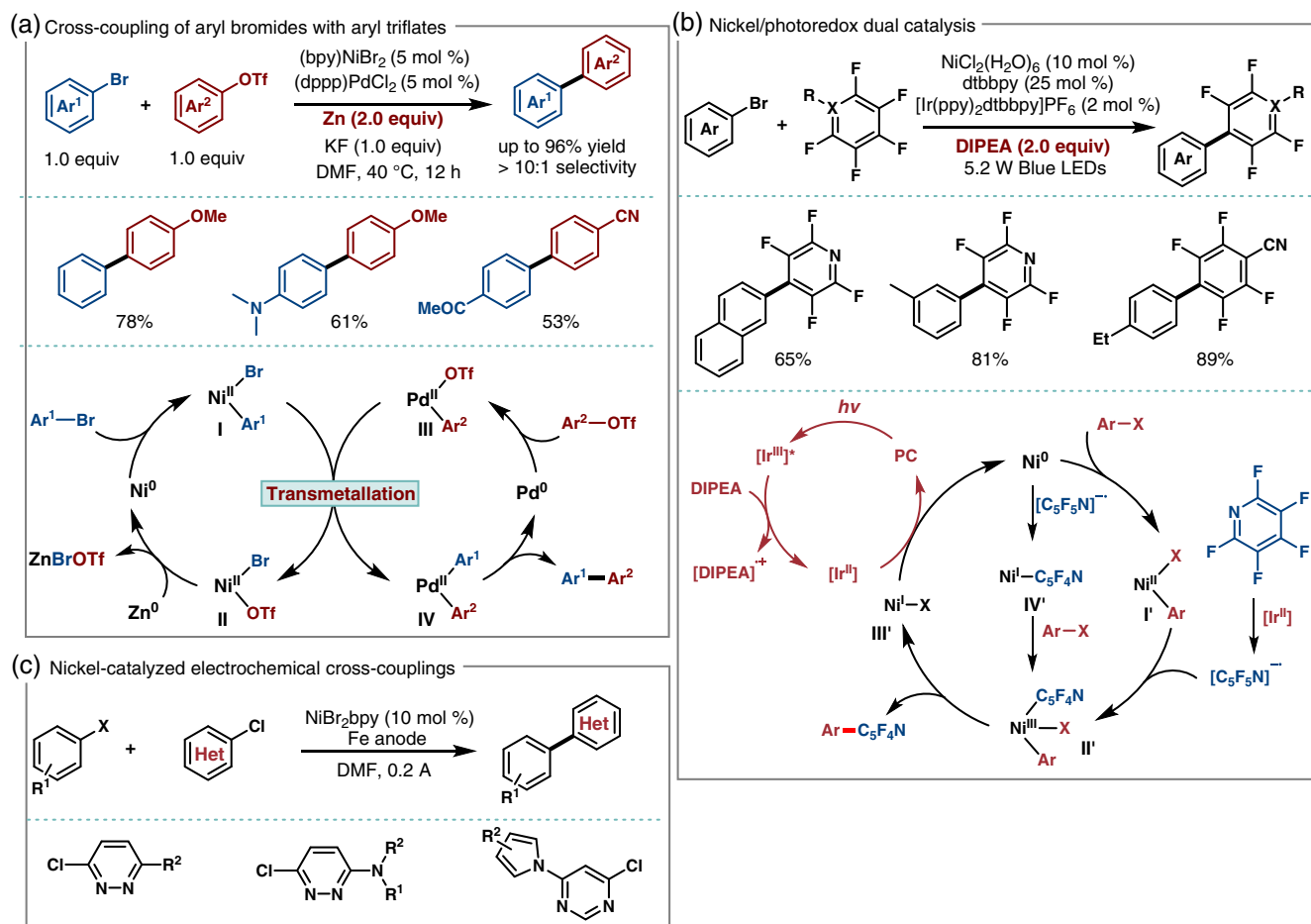
Aryl-Aryl Reductive Cross-Couplings

In comparison to cross-electrophile coupling for the construction of C(sp²)-C(sp³) bonds, the corresponding C(sp²)-C(sp²) bond formations are more challenging as a result of the subtle difference between the two aryl electrophiles. In 2008, Amatore and Gosmini⁸⁹ reported a cobalt-catalyzed reductive cross-coupling of two aryl

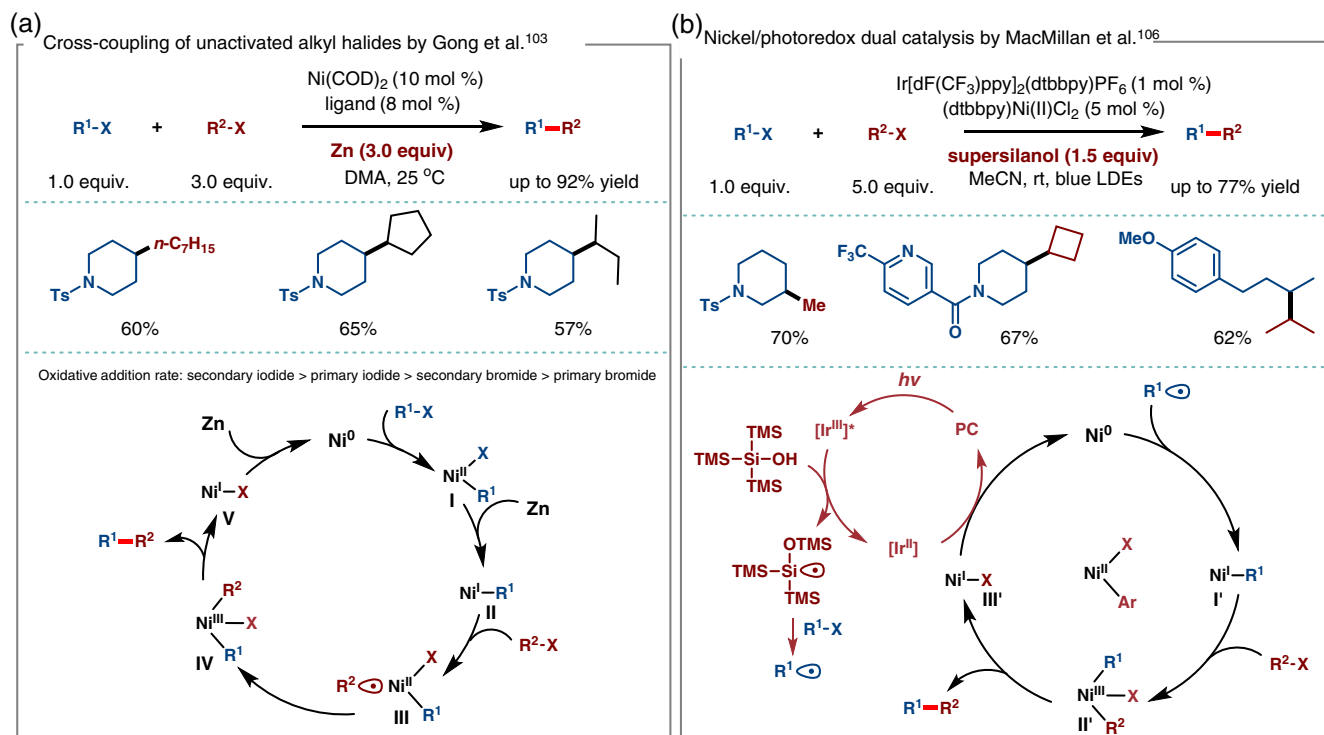
electrophiles for the synthesis of unsymmetrical biaryl compounds in the presence of Mn as the reductant. The key to success was the different reactivity profiles of two aryl electrophiles, which allowed the selective oxidative addition of these species. Later, this concept was extended to nickel-catalyzed cross-coupling of aryl halides with 2-halopyridine by the same group.⁹⁰ In 2015, Weix and co-workers⁹¹ developed a different method to construct C(sp²)-C(sp²) bonds, wherein the selectivities were controlled by the catalysts rather than the electronic properties of the substrates as previous reports (Scheme 12a). Mechanistically, aryl bromides undergo selective oxidative addition with Ni(O) to give Ni(II) species **I**. At the same time, aryl triflates react exclusively with Pd(O) to form Pd(II) species **III**. The transmetalation then occurs between Ni(II) and Pd(II) complex, leading to the key Ar¹-Pd(II)-Ar² intermediate **IV**, which allows for the reductive elimination to give the asymmetrical biaryls. Notably, the additive potassium fluoride (KF) is crucial to achieving high selectivity, presumably by improving the selectivity of the palladium catalyst for aryl triflate over aryl bromide. Continued efforts by Olivares and Weix⁹² and other groups expanded this method for the coupling of vinyl bromides with vinyl triflates, aryl chlorides with aryl triflates,⁹³ aryl

triflates with aryl tosylates,⁹⁴ aryl chlorides with ortho-fluoro-substituted aromatic amides,⁹⁵ aryl iodides with difluoromethyl 2-pyridyl sulfone,⁹⁶ aryl bromides, and 2,2-difluorovinyl tosylate.⁹⁷

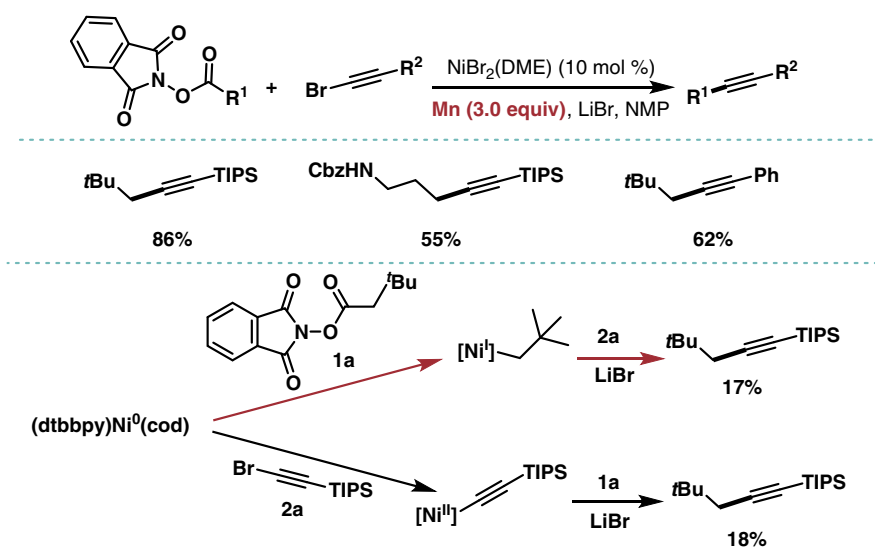
Very recently, Rueping and co-workers⁹⁸ realized the reductive cross-coupling of polyfluorinated arenes with aryl halides using nickel/photoredox dual catalysis (Scheme 12b). This protocol opens up a new entry to multifluorinated biaryls. The reaction process starts with the oxidative addition of aryl halides with Ni(O) generating Ni(II) species **I'**. Concomitantly, the [C₅F₅N][−] is generated via a SET process of Ir(II) to C₅F₅N, which can be trapped with Ni(II) to give Ni(III) intermediate **II'**. The following reductive elimination affords the desired product and Ni(I) species **III'**, which can be reduced by Ir(II). Also, a Ni(O)/Ni(I)/Ni(III)/Ni(I) cycle is possible, a pathway that involves the trapping of Ni(O) and selective addition of Ni(I)-C₅F₄N (**IV'**) to the aryl halides. Besides these achievements, the nickel-catalyzed electrochemical method was also applied to aryl-heteroaryl bond formation reactions. In this regard, Léonel and co-workers^{99–102} reported several electroreductive nickel-catalyzed cross-coupling reactions of aryl halides and heteroaryls, such as 3-chloro-6-methoxypyridazines,



Scheme 12 | (a–c) Nickel-catalyzed aryl–aryl reductive cross-couplings via different methods.



Scheme 13 | (a and b) Nickel-catalyzed alkyl-alkyl reductive cross-couplings via different methods.



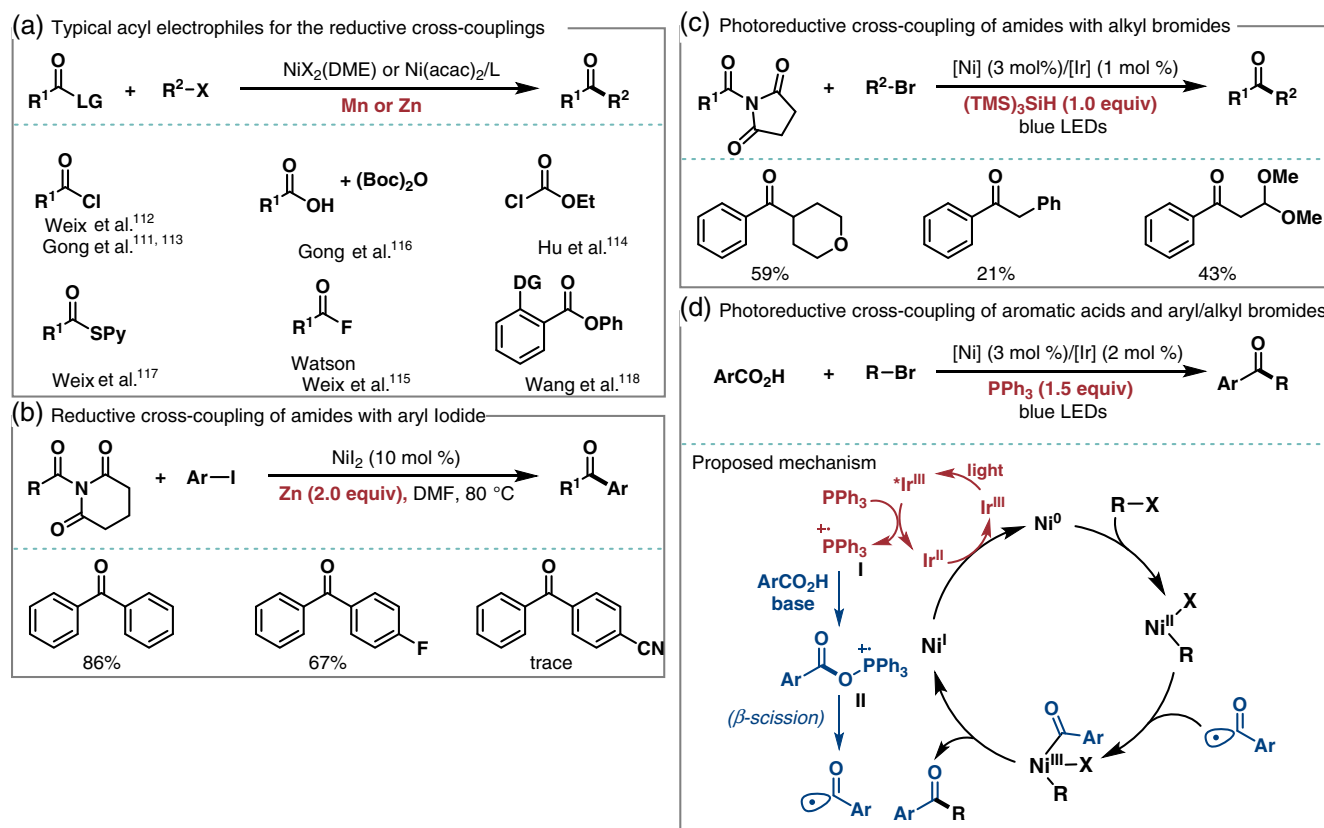
Scheme 14 | Nickel-catalyzed reductive alkynylation of NHP esters.

3-amino-6-chloropyridazines, and chloropyrimidines (Scheme 12c).

Alkyl-Alkyl Reductive Cross-Couplings

As discussed above, the reductive cross-coupling construction of $\text{C}(\text{sp}^2)-\text{C}(\text{sp}^3)$ and $\text{C}(\text{sp}^2)-\text{C}(\text{sp}^2)$ bonds can

be achieved by taking advantage of the different reactivity profiles between the alkyl and aryl reaction partners or two aryl electrophiles. However, the corresponding cross-coupling synthesis of $\text{C}(\text{sp}^3)-\text{C}(\text{sp}^3)$ bonds is even more challenging due to the difficulty of suppressing the competing homo-coupling pathways. In 2011, Gong and co-workers¹⁰³ first reported the nickel-catalyzed reductive cross-coupling of two different alkyl halides using 2,6-bis(4,5-dihydrooxazol-2-yl)pyridine (pybox) as



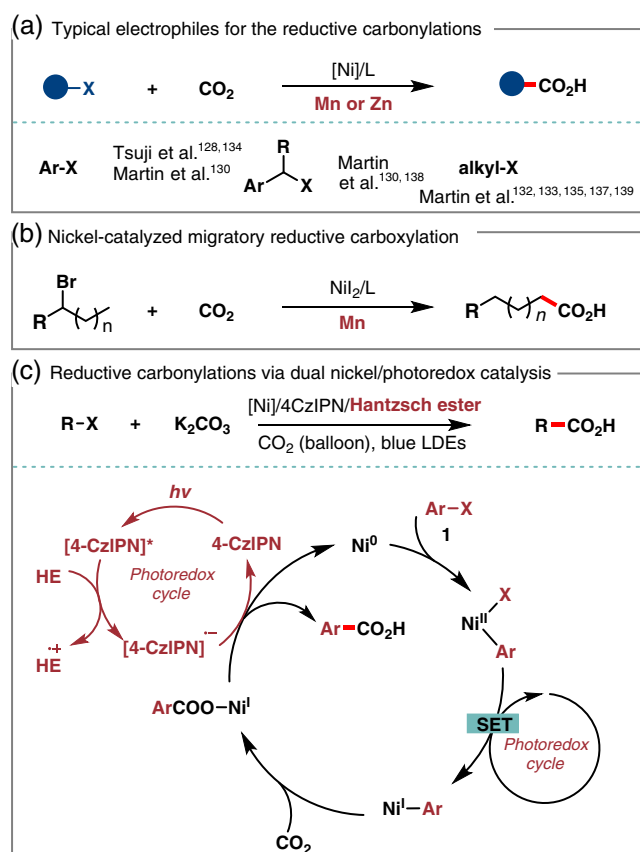
Scheme 15 | (a–d) Nickel-catalyzed reductive acyl cross-couplings via different methods.

the ligand and Zn as the reductant. The reaction tolerated various functional groups. A wide range of alkyl halides underwent the cross-couplings in moderate to good yields, affording the desired products. Continuing their efforts, further improvement in chemoselectivity was achieved by using (Bpin)₂ (Bis(pinacolato)diboron) as the reductant.¹⁰⁴ A proposed mechanism for this transformation is provided in Scheme 13a, where the oxidative addition of R¹-X to Ni(0) generates the Ni(II) intermediate **I** and is subsequently reduced by Zn to give Ni(I)-R¹ intermediate **II**. A SET or halide abstraction process would then occur to generate the alkyl radical, followed by the formation of Ni(III) species **IV**. Finally, the reductive elimination of Ni(III) species delivers the desired product and Ni(I) species **V**, which is reduced by Zn to regenerate Ni(0). Despite this progress, the development of a general method allowing for the reductive cross-coupling of two secondary alkyl halides is still a challenge. A related approach was developed by Wang and co-workers¹⁰⁵ for the assembly of primary aliphatic fluorides with bromofluoromethane and alkyl bromides. Recently, MacMillan and co-workers¹⁰⁶ successfully addressed this issue via nickel/photoredox dual catalysis (Scheme 13b). The reaction had a wide substrate scope, including methyl tosylates, cyclic alkyl bromides, and secondary alkyl halides.¹⁰⁶ The reaction is proposed to proceed by SET oxidation of the

(TMS)₃SiOH for the formation of silyl radicals, which can produce the key alkyl radical via halogen-atom abstraction from alkyl bromides. Subsequently, the alkyl radical reacts with a Ni(0) to generate Ni(I)-R **I'**. The following oxidative addition and reductive elimination process will give the cross-coupling product and Ni(I) species **III'**, which is reduced by the reduced state of Ir(II) to close both catalytic cycles.

Alkyl-Alkynyl Reductive Cross-Couplings

In comparison with above-mentioned C-C bond formations, the corresponding alkyl-alkynyl reductive cross-couplings remain unexplored possibly due to the difficulty in controlling the selectivity of bromoalkynes and suppressing the side reactions. In addition, the prosperity of Sonogashira coupling makes this type of reaction less relevant. In 2017, Weix and co-workers¹⁰⁷ reported the first nickel-catalyzed cross-electrophile coupling of alkynyl bromides with NHPI esters for the construction of C(sp³)-C(sp) bonds (Scheme 14). Mechanism studies indicated that the reaction proceeds through nickel-mediated bond formation, rather than involving an alkynylmanganese reagent.



Scheme 16 | (a–c) Nickel-catalyzed reductive carboxylations via different methods.

Reductive Acylation and Carboxylation Reactions

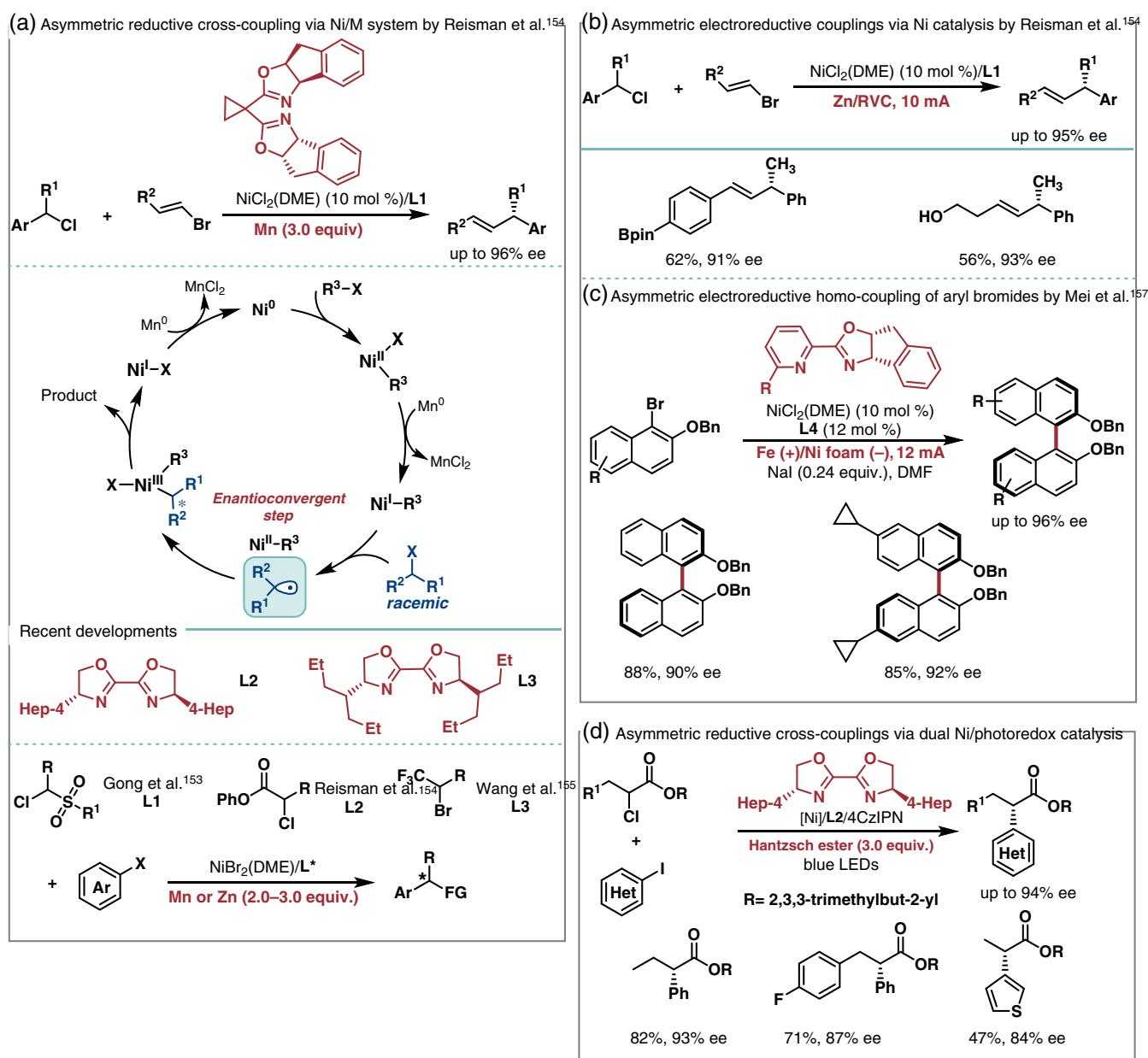
Ketones have attracted great attention due to their wide applications in natural products and pharmaceutical molecules and materials.^{108–110} This has resulted in the development of many synthetic methods. In this regard, nickel-catalyzed reductive ketone formation has become a powerful tool to construct asymmetric ketones (Scheme 15a), and a series of reductive carbonylation reactions have been developed with several acid derivatives, such as acid chlorides,^{111–114} acid fluorides,¹¹⁵ anhydrides,¹¹⁶ thioesters,¹¹⁷ phenolic esters,¹¹⁸ and in situ activated acids.^{119,120} In recent years, tremendous progress was made to utilize stable amides in cross-coupling reactions to synthesize ketones.¹²¹ In 2017, Han and co-workers¹²² reported the nickel-catalyzed reductive cross-coupling of amides with aryl iodides in the presence of Zn as the reductant (Scheme 15b). This process was compatible with a wide range of aryl iodides and provided diaryl ketones in moderate to very good yields. The mechanistic studies suggested that an aryl radical is generated from aryl iodides via the reduction of Ni(II) species. A related approach was developed by Li and co-workers¹²³ for the reductive acylation

of acylimidazoles with aryl and alkyl bromides in the presence of Zn as the reductant. Interestingly, Amgoune and co-workers¹²⁴ recently realized a nickel/photo dual catalyzed reductive cross-coupling of amides and alkyl bromides for the synthesis of alkyl ketones (Scheme 15c). The reaction proceeded well with various functionalities on the phenyl ring of the amides. Similarly, good functional group tolerance was also observed with respect to the substituents on primary and secondary alkyl bromides, including esters, nitriles, amines, ethers, and acetals. Similar to previous reports, the alkyl radical was generated through the halogen-atom abstraction of (TMS)₃Si radical. A much simpler process initiated by nickel/photoredox dual catalysis is shown in Scheme 15d. Xie and co-workers¹²⁵ described a cross-electrophile coupling of aromatic acids and aryl/alkyl bromides. This protocol opens a new avenue to ketone synthesis, thereby avoiding the preparation of acid derivatives. In this process, triphenylphosphine is used for the SET reduction of photoexcited catalyst *Ir^{III}. The phosphoranyl radical intermediate **II** is then generated from the reaction of triphenylphosphine radical cation **I** and carboxylate anion. The following β -scission of phosphoranyl radical intermediate gives the acyl radical.¹²⁵

Carbon dioxide (CO₂), which is an abundant and renewable carbon resource, has been extensively used as a valuable C1 building block in various organic syntheses. Nickel-catalyzed electroreductive cross-coupling of aryl halides and CO₂ was reported in the early 1980s.^{126,127} Recently, numerous efforts have been focused on CO₂ fixation via reductive nickel catalysis. In this regard, a breakthrough was not achieved until very recently. In 2012, Tsuji and co-workers¹²⁸ reported a nickel-catalyzed carboxylation of aryl and vinyl chlorides under a CO₂ pressure of 1 atm at room temperature in the presence of Mn as a reducing agent (Scheme 16a). Subsequent work established the nickel-catalyzed reductive carboxylations with a wide variety of electrophiles. Under nickel/Mn or Zn conditions, aryl and vinyl triflates, benzyl, alkyl pseudo-electrophiles, alkenyl and alkynyl compounds have been utilized for the synthesis of various carboxylates.^{129–141} Notably, nickel-catalyzed reductive migratory cross-coupling of alkyl bromides with CO₂ was also realized by Martin and co-workers¹⁴² (Scheme 16b). Recently, König and co-workers¹⁴³ leveraged a nickel/photoredox dual catalysis for the reductive carboxylation of aryl and alkyl bromides/triflates (Scheme 16c). In this case, the reaction featured a broad substrate scope and functional group tolerance in the presence of K₂CO₃ as the source of CO₂ and HE as the reductant.

Asymmetric Reductive Cross-Couplings

Metal-catalyzed cross-coupling of secondary alkyl electrophile is a direct and step-economic synthetic strategy

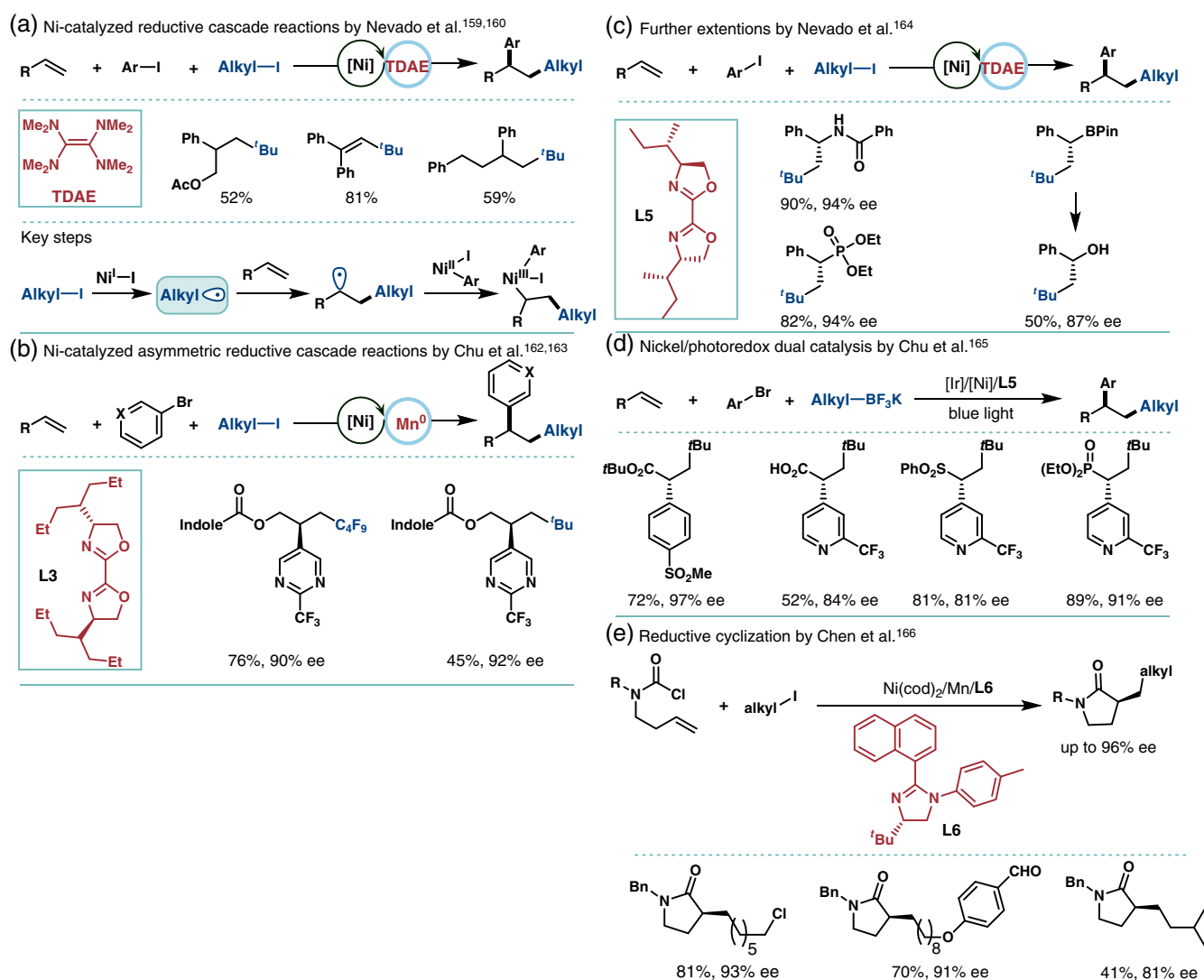


Scheme 17 | (a–d) Nickel-catalyzed asymmetric reductive cross-coupling via different methods.

for the asymmetric construction of C–C bonds,¹⁴⁴ but challenges remain regarding selectivity and enantioconvergence of racemic starting materials. The formation of alkyl radicals from racemic alkyl electrophiles provide a powerful entry to scramble the stereocenter.¹⁴⁵ In this context, Reisman and co-workers¹⁴⁶ developed several enantioconvergent cross-coupling of racemic alkyl electrophiles in the presence of stoichiometric amounts of metallic reducing agents, including acylation, vinylation, arylation of racemic benzylic chlorides, and arylation of racemic α -chloronitriles.^{147–152} These results led to the further development of the arylation of α -chlorosulfones, α -chloroesters, and trifluoromethylated alkyl bromide by the research groups of Gong,¹⁵³ Reisman,¹⁵⁴ and Wang.¹⁵⁵

Mechanistically, enantioconvergence is achieved via the activation of an alkyl electrophile to form the alkyl radical (Scheme 17a).

Furthermore, by taking the beneficial effects of electrochemistry in reductive cross-couplings, the same group realized the first asymmetric electroreductive cross-coupling of alkenyl and benzyl halides in an undivided cell (Scheme 17b).¹⁵⁶ This concept was also used by Mei and co-workers,¹⁵⁷ who disclosed the electrochemical reductive homocoupling of aryl bromides, affording enantioenriched axially chiral biaryls in good yield and enantioselectivities (Scheme 17c). Notably, Mao and co-workers¹⁵⁸ have shown the application of nickel/photoredox dual catalysis in asymmetric



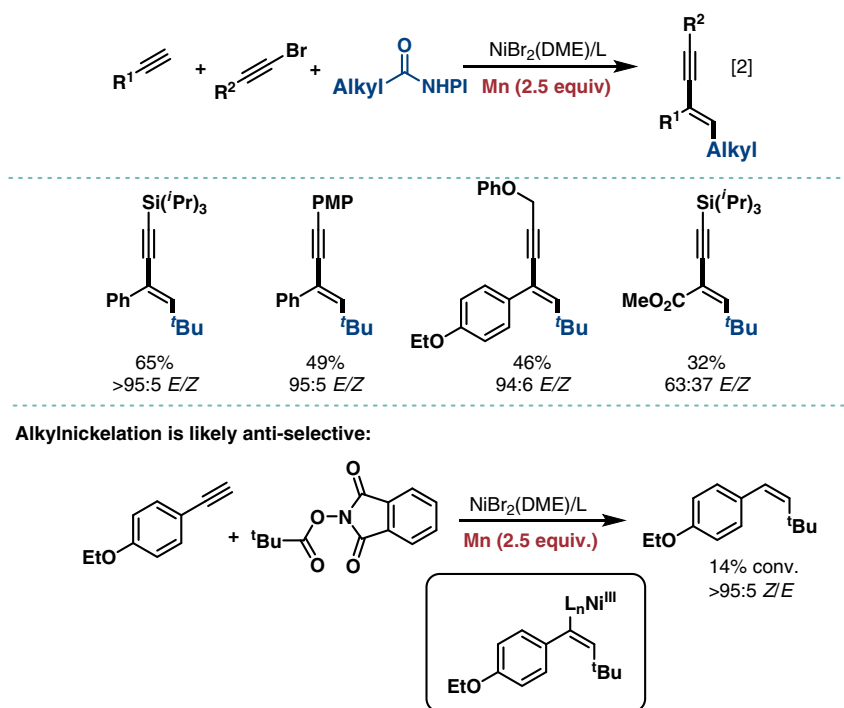
Scheme 18 | (a–e) Nickel-catalyzed reductive cascade reactions.

reductive cross-coupling of α -chloro esters and aryl iodides (Scheme 17d). This coupling, in the presence of HE as the reductant, afforded α -aryl esters in generally good yields and very good enantioselectivities.¹⁵⁸

Miscellaneous

In a significant addition to the reductive cross-couplings, Nevado and co-workers^{159,160} successfully developed a nickel-catalyzed three-component 1,2-alkylarylation of inactivated alkenes (Scheme 18a). The reaction proceeded smoothly using tetrakis(dimethylamino)ethylene (TDAE) as the organic reducing agent, with various alkyl iodides, aryl iodides, and inactivated alkenes. Mechanically, the alkyl radical is generated via SET or halogen-atom abstraction of the low-valent nickel intermediate. Subsequent radical addition to alkenes and the nickel-catalyzed radical coupling give the desired products. This concept was also developed by Chu

and co-workers,¹⁶¹ who disclosed the nickel-catalyzed three-component carboacylation of inactivated alkenes with acyl chlorides and perfluoroalkyl halides in the presence of Mn as the reductant. Very recently, Chu and co-workers^{162,163} developed an enantioselective version of this three-component reductive dicarbofunctionalization process using (*R,R*)-4-heptyl-BiOX **L3** as the chiral ligand and Mn as the reductant (Scheme 18b). Unfortunately, the ester directing group of alkenes was necessary to achieve high enantioselectivities.^{162,163} Remarkably, Nevado and co-workers¹⁶⁴ further extended this chemistry by using vinyl amides, vinyl boranes, and vinyl phosphates as suitable alkenes (Scheme 18c). Based on previous developments, Chu and co-workers¹⁶⁵ described an interesting visible light-promoted nickel/photoredox dual strategy for the construction of enantioenriched β -alkyl- α -arylated carbonyls, phosphonates, and sulfones by employing a series of different olefins (Scheme 18d). Most recently, Chen and co-workers¹⁶⁶ designed a new



chiral 8-quinoline imidazoline ligand for the reductive cyclization of monosubstituted alkenes. This sequence, in the presence of $Ni(cod)_2/L5$, a stoichiometric amount of Mn , afforded the tertiary α -alkylated pyrrolidinone in very good enantioselectivities (Scheme 18e).¹⁶⁶

The strategy of reductive cascade activation modes was also successfully expanded to reactions with alkynes, alkynyl bromides, and NHPI esters by Koh and co-workers.¹⁶⁷ In this case, the cascade sequence furnished the corresponding 1,3-enynes in moderate to good yields with good to excellent E/Z ratio, which presumably originated from the alkylnickelation intermediate (eq 2).¹⁶⁷

Summary

Nickel-catalyzed reductive cross-couplings provide an efficient and economical way for the construction of C–C bonds using abundant and stable electrophiles as coupling partners. However, the requirement of a stoichiometric reductant can complicate scale-up and generate excess stoichiometric metal waste. Taking advantage of the development of synthetic techniques especially electro- and photochemistry, cross-electrophile coupling has become more environmentally friendly for selective C–C bond formation without using metal powders. The newly developed environmentally friendly techniques, which enable chemists to create new activation modes, will be one of the future directions in this area. An important step in reductive cross-coupling is to reduce the high-valent nickel intermediate and

regenerate the active nickel catalyst. On the basis of the types of reducing agents, three reaction protocols have been used in cross-electrophile couplings: (1) Traditional strategy-stoichiometric metallic reducing agents are employed for the reduction of $Ni(II)$ and $Ni(I)$ species; (2) Photochemistry in which a photoredox catalyst and an organic reductant are used for generation of alkyl radicals and regeneration of the nickel catalyst; (3) Electrochemistry in which direct cathodic reduction is used for the reduction of $Ni(II)$ and $Ni(I)$ species and the generation of organic radicals. The development of photo- and electroreductive cross-coupling is still in its infancy. In the case of nickel-catalyzed reductive cross-couplings, future progress will result from efforts to develop $C(sp^2)$ – $C(sp^2)$ and $C(sp^3)$ – $C(sp^3)$ bond formations and more general enantioselective approaches. Importantly, a better understanding of the reaction mechanism, in particular with respect to the photo- and electrochemical reactions, is required to design more stable, efficient, and scalable catalyst systems for these valuable cross-electrophile couplings.

Conflict of Interest

There is no conflict of interest to report.

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